

Let $\ell(\pi) = S \log(\pi) + (n - S) \log(1 - \pi)$ for $0 < \pi < 1$, where S and n (with $0 \leq S \leq n$) are fixed numerical constants.

Find $\frac{d\ell(\pi)}{d\pi}$ and $\frac{d^2\ell(\pi)}{d\pi^2}$.